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Social Media Influence on Tobacco Policy Support: The Role of Cancer Worry and Smoking Status in a Moderated Mediation Framework

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ABSTRACT

This study examined whether social media use for health purposes is associated with support for tobacco control policies among individuals with a family history of cancer. We employed a moderated mediation framework and analyzed the Health Information National Trends Survey (HINTS 5, Cycle 4; $N = 2,671$) data, a nationally representative sample collected from the United States. We found that cancer worry mediated 4.6% of the total effect, with smoking status positively moderating this pathway. While nonsmokers showed higher baseline support, smokers exhibited steeper increases in policy backing as cancer worry intensified. Results aligned with the stimulus–organism–response (S-O-R) model, suggesting targeted social media messaging could enhance tobacco policy advocacy, particularly for smokers with familial cancer risks. The study offers theoretical contributions to public health and health communication, as well as practical implications for policymakers and the general public.

Introduction

Cancer stands as a formidable global health challenge, claiming a substantial toll on human lives. With recent estimates indicating the burden continues to rise, the year 2020 alone witnessed nearly 10 million deaths attributed to cancer worldwide (World Health Organization [WHO], 2026a). Among the cancer risk factors, smoking remains a primary culprit, responsible for approximately 7 million deaths globally each year (WHO, 2026b). Smoking significantly harms health and is a primary cause of four major diseases, including lung cancer, heart disease, stroke, and chronic obstructive pulmonary disease (Lee, 2018).

Given the strong association between smoking and cancer risk, tobacco control policies have emerged as vital tools for effective interventions and early detection, demonstrating effectiveness in cancer prevention (Bialous et al., 2003; Gredner et al., 2021; Munthali et al., 2021). For example, policies like raising tobacco taxes (Chaloupka et al., 2012), enforcing users' age limits (Dobbs et al., 2023), setting smoke-free zones (Semple et al., 2022), and displaying warning labels all aim to reduce tobacco harm (Strong et al., 2021). With the efforts, we have observed a steady decline in cigarette consumption and a salient reduction in lung cancer and other smoking-related health problems (Feng et al., 2019; Paoletti et al., 2012). Such effectiveness has gained wide public support: research indicates that public opinion generally favors stricter tobacco control (Henderson et al., 2022; Mumford et al., 2017), such as smoking restrictions and bans on tobacco advertising (Bakhturidze et al., 2013). Public support was highest in places with smoke-free policies, especially where children were most

likely present (Rosenberg et al., 2012). Recent surveys from various countries, including England, Canada, Australia, and the United States, have demonstrated widespread advocacy for tobacco restrictions among adults (Hawkins et al., 2020).

Since public attitudes toward tobacco and relevant policies are crucial foundations for legislation and law enforcement, scholars care about the factors that shape certain policy orientations. Among the channels where people form public opinion toward tobacco, social media has gained scholarly attention (Donaldson et al., 2024). Social media is defined as “the use of the Internet to connect with others through social networks, such as Facebook, LinkedIn, Twitter, and other similar platforms” (Li et al., 2024, p. 3). In the past decades, social media has become an influential source of health-related information (Vance et al., 2009; Y. Zhao & Zhang, 2017) and exerts an impact on tobacco use through exposure to tobacco-related content. Importantly, greater social media use leads to more exposure to all forms of tobacco content, including both promotional material (Cavazos-Rehg et al., 2021) and public health information (Huang et al., 2016; L. Zhang et al., 2024). Tobacco-related content on social media is heterogeneous. We can categorize the information into three groups based on their orientation: (1) promotional content that encourages tobacco consumption; (2) critical content that is anti-tobacco; and (3) neutral content that does not have a clear inclination.

The first category of “promotional content” includes the direct and indirect advertising by tobacco companies, as well as sponsored content by influencers that portray tobacco use in a positive light (Liang et al., 2015). Such content often

circumvents platform policies (e.g., X, formerly Twitter) and targets populations at greater risk, including youth (Kong, Laestadius, et al., 2024). Exposure to promotional content heightens brand awareness, product appeal, and susceptibility to tobacco use (Donaldson et al., 2022), which may undermine support for tobacco control policies by normalizing tobacco use and reducing risk perceptions.

The second category is against the usage of tobacco. Social media has been associated with the dissemination of anti-tobacco messages and fostering public attitudes toward tobacco restrictions. Campaigns utilizing social media have been adopted to reach a broad audience and influence public health perceptions on tobacco use (Kong, Blake, et al., 2024), with initiatives like the Truth Initiative demonstrating significant success (Duke, Alexander, et al., 2015; Farrelly et al., 2009; Vallone et al., 2018). Studies increasingly report positive associations between social media use and support for tobacco restrictions. For example, Hefler et al. (2013) found three cases of how social media use facilitated the advancement of tobacco control. Similarly, Hamill et al. (2015) estimated the leveraged communication channels like Facebook and online advertising to promote tobacco control policies.

The third category would be neutral or ambiguous content about tobacco on social media. For instance, news reports, personal testimonials without clear advocacy, and general discussions about tobacco exist on social media. More broadly, different types of tobacco-related content have been shown to elicit varying levels of user engagement (Chu et al., 2017), and exposure to e-cigarette marketing across media platforms has been found to differ by demographic characteristics (Emery et al., 2014).

Beyond influencing public attitudes toward tobacco, social media is also a crucial health information source for individuals, particularly among those facing health challenges. For instance, people with a cancer diagnosis and their family are particularly engaged in using social media as an important source of health communication (Braun et al., 2019). Furthermore, people with family cancer history often exhibit certain characteristics as family cancer history associates with one's genetic susceptibilities, cancer risk perception, and health behaviors (Yoon et al., 2002). People with family histories of cancer usually possess stronger risk perception and cancer worry. For instance, Molina et al. (2015) indicated that women with a family history of cancer often undergo regular health checks due to worry and have stronger perceptions of their future cancer risk. Additionally, Seven et al. (2018) investigated that women's understanding of the hereditary factors of breast cancer and risk reduction strategies was moderate, yet the majority experience moderate to high cancer worry and are concerned about the possibility of developing breast cancer. These individuals with a family cancer history deserve greater attention. They bear both the emotional burden of their relatives' cancer experience and often practical caregiving sacrifices and develop distinct perceptions of carcinogens and cancer prevention policies.

Exposure to cancer-related content on social media can amplify health anxiety (Liu, 2020); this effect is particularly pronounced for cancer worry, where actively seeking cancer information online heightens emotional distress (Hoffman-

Goetz et al., 2009). Therefore, exposure to health-specific media content could augment cancer worry (Jung, 2014). The act of seeking cancer-related information on social media will highlight family cancer history as a substantial cancer risk factor; consequently, this boosted awareness of family health risk could enhance both the perceived risk of cancer and the levels of cancer worry (Butterworth et al., 2006).

Elevated cancer worry encourages inclination toward health-protective behaviors (Ferrer et al., 2013). Empirical research suggests that, for both smokers and nonsmokers, cancer has specific causes and can be prevented, and this finding is a strong indicator of support for workplace smoking control policies (Brown et al., 1988). Y. Chen and Yang (2017) revealed that cancer worry positively influences the perceived advantages of quitting smoking. Individuals who perceived greater cigarette-related harm and held the belief that smoking causes cancer were more likely to have quit smoking (Land et al., 2023). Furthermore, cancer worry could positively affect health-protective behavior, such as tobacco control (Li et al., 2024).

However, the relationship between cancer worry and support for tobacco restrictions is unlikely to be uniform across all individuals. This relationship varies significantly based on current smoking behavior. Indeed, smoking status was a potential moderator identified by prior research such as the study by Reis and Postolache (2019), which indicates that smokers and nonsmokers perceive tobacco risks differently. Though the current literature does not explicitly address the specific moderating role played by smoking status in the relationship between cancer worry and support for tobacco control policies, the insights suggest it is a promising mechanism to explore. For instance, scholars indicate that variations in smoking status dictate the impact of perceived risk on the family history of lung cancer, a dynamic more pronounced among nonsmokers (L. Chen & Kaphingst, 2010). Moreover, smokers and nonsmokers may respond differently to cancer worry. Nonsmokers view cancer worry as a cue for health-protective behaviors, reinforcing support for tobacco control (Boderie et al., 2023; Finney Rutten et al., 2011). In contrast, smokers may experience cognitive dissonance and downplay cancer risks (Dillard et al., 2006; Fotuhi et al., 2013). These studies therefore prompt a focused investigation into the moderating role of smoking status in the relationship between cancer worry and support for tobacco restrictions across smokers and nonsmokers.

The stimulus-organism-response (S-O-R) model, developed by Mehrabian and Russell (Mehrabian & Russell, 1974), offers a framework for understanding how external stimuli shape emotional states and influence outcome behaviors. Within the S-O-R framework, exposure to health-related social media content constitutes the stimulus (S), operationalized as the external factor triggering internal processes. This stimulus evokes cancer worry as the mediating organism (O)—an affective state manifesting as fear, concern, or heightened risk perception regarding tobacco-related health threats. Subsequently, support for tobacco control policies emerges as the behavioral response (R). This operationalization aligns with empirical evidence demonstrating that social media exposure initiates emotional responses (Tsay et al., 2020; X. Zhang

et al., 2023), which in turn shape health policy attitudes (Fakfare & Sangpikul, 2024; Fan et al., 2024).

This framework has been widely adopted by multiple fields, especially in public health research. For instance, Fan et al. (2024) found that lower levels of health literacy are associated with greater resistance to applying electronic health services in Nigeria. Similarly, Sultan et al. (2021) explored the influence of marketing communication channels and the core values of organic food on consumers' behavioral intentions toward organic products. Another study conducted by Fu et al. (2021) confirmed that overuse of smartphones may trigger health issues such as insomnia, nomophobia, and poor eyesight in students, which may negatively impact their academic performance. Applying this framework, the present study addresses a critical gap by examining whether social media use for health, mediated by cancer worry and moderated by smoking status, is associated with policy support within a population with a family cancer history (defined as any of the first- or second-degree biological relatives ever diagnosed with any type of cancer), offering insights for targeted public health communication strategies.

Despite extensive tobacco-control research employing longitudinal and experimental designs (Duke, Davis, et al., 2015; Farrelly et al., 2009), the mechanism linking media exposure to policy support remains underspecified. While smoking behaviors among people with cancer have received attention (Tyc et al., 2013), their relatives' policy attitudes remain unexamined. Moreover, whether cancer worry differentially predicts policy support for smokers versus nonsmokers, among those sharing a family cancer history, has never been tested. The current study offers three contributions: (1) theorizing cancer worry as an affective mediator within the S-O-R framework, a mechanism not previously modeled; (2) targeting an understudied population with elevated risk (with a family cancer history) rather than general samples; (3) testing smoking status as a moderator of the worry-policy relationship.

Research questions and hypotheses

Based on statistical significance and non-significance, traditional hypothesis testing often over-relies on dichotomy and categorical differentiation (Amrhein et al., 2019). For mediation and moderation analyses, X. Zhao et al. (2024) emphasize the typology and effect size interpretation. Therefore, we draw our research questions as follows: What type of mediation best describes the effect of social media use for health on support for control policies of tobacco through cancer worry? What is the effect size variance in the relationship between social media use for health and support for tobacco control policies mediated through cancer worry and moderated by smoking status? Based on the relevant literature and the above questions that we plan to investigate, here are our research hypotheses (conceptual model in Figure 1):

H1: Social media use for health is positively associated with support for tobacco control policies via its direct effect (*d* path).

H2: Social media has a positive indirect effect on support for tobacco control policies through cancer worry. That is, social media use for health is positively associated with cancer worry

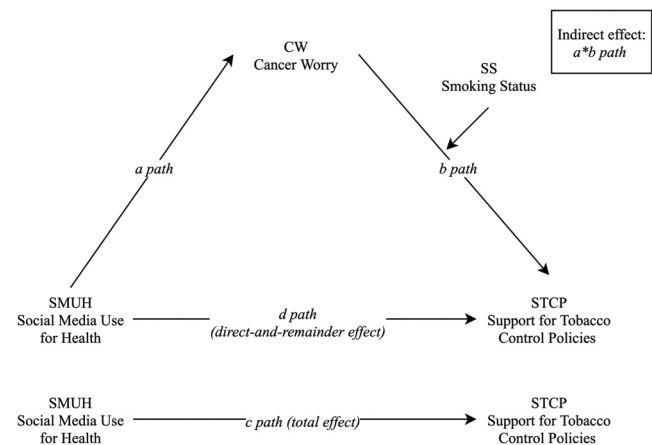


Figure 1. Conceptual model linking social media use for health to tobacco policy support through cancer worry, moderated by smoking status.

(*a* path), and cancer worry positively correlates with support for tobacco restrictions (*b* path).

H3: Social media use for health is positively associated with support for control policies of tobacco, which means the total effect (*c* path).

H4: Smoking status positively moderates the relationship between cancer worry and support for tobacco control policies.

H5: Smoking status positively moderates the indirect path between social media use for health and support for tobacco control policies through cancer worry.

Method

Data source

This study utilizes data from the Health Information National Trends Survey (HINTS), a cross-sectional survey project sponsored by the National Cancer Institute. The HINTS project is a publicly available data source that is approved by the Westat Institutional Review Board. It also received a Not Human Subjects Research determination from the NIH Office of Human Subjects Research, ensuring compliance with ethical standards in human subjects research. Therefore, our analysis of HINTS does not raise ethical concerns and causes no harm to human participants. More details about HINTS data could be extracted from the HINTS website (Winston, 2021).

This study specifically selects data from the HINTS 5, Cycle 4 (2020), which employed a two-stage sampling design, and the data collection occurred via mail between February 24 and June 15 2020, with a response rate of 36.7%. The survey weights accounted for differential selection probabilities, non-response, and raking to population benchmarks from the 2018 American Community Survey and health characteristics. Replicate weights enabled variance estimation for complex analyses. The present study focuses on respondents with a family cancer history. The final sample consists of 2,671 individuals drawn from 3,865 valid respondents. This subgroup provides valuable insights into the relationship between

social media, cancer worry, and health policy support. We applied listwise deletion to handle missing or invalid responses, excluding incomplete records from the regression analyses. This resulted in the exclusion of 337 individuals, an acceptable level of loss (12.61% of 2,671), thus retaining the majority of the original sample. The analytic sample for the moderated mediation model therefore consists of 2,334 complete observations.

Measurements

The dependent variable, support for tobacco control policies (STCP), was measured by five questions (Kaufman et al., 2022; Shete et al., 2021). Two of them are about cigarettes; their wordings read as follows: “To what extent would you support or oppose the following measures related to cigarettes? 1. Just like with violence and sex, movies with cigarette smoking should be rated ‘R’ to protect children and youth from seeing cigarette smoking in movies. 2. Cigarette packs should be required to have warning labels that use both images and words to show the negative health effects of smoking.” Three focus on tobacco: “To what extent would you support or oppose the following measures related to all tobacco products, including cigarettes, e-cigarettes, smokeless tobacco, hookah, and cigars? 1. Stores should be required to keep tobacco products out of customers’ view at the checkout counter. 2. Stores should be required to keep advertisements for tobacco products away from cash registers and out of windows. 3. Tobacco products should not be advertised on social media.” Possible responses ranged from 1 = *strongly oppose* to 5 = *strongly support*. A composite variable was computed by averaging the five items ($M = 3.79$, $SD = 0.92$, Cronbach’s $\alpha = 0.89$). The coding is intuitive: the higher the value, the greater the support for tobacco control.

Social media use for health (SMUH) (Chou et al., 2021; Huo et al., 2019) was measured by a set of questions derived from HINTS items, which captured a basic breadth of engagement but lacked granularity regarding frequency, intensity, duration, or platform-specific nuances of use. This binary operationalization (used/did not use) is a simplification of a complex behavior. However, it is appropriate for establishing foundational associations and aligning with prior work utilizing HINTS data (Lama et al., 2022; Onyeaka et al., 2023). The assessment consisted of the following three questions: “In the past 12 months, have you used the Internet for any of the following reasons? 1. To share health information on social networking sites, such as Facebook or Twitter; 2. To participate in an online forum or support group for people with a similar health or medical issue; or 3. To watch a health-related video on YouTube.” Social media use for health was coded as 1 if respondents reported using the internet for any of the above reasons. A SMUH measurement index was created by calculating the mean of the dichotomous items directly ($M = 0.20$, $SD = 0.27$). A greater value on the SMUH index indicates greater engagement with social media applications.

Cancer worry (CW) was assessed by measuring a respondent’s level of concern regarding the possibility of developing cancer (Liu et al., 2023). Participants were asked,

“How worried are you about getting cancer?” and responded on a five-point scale ranging from 1 = not at all to 5 = extremely ($M = 2.89$, $SD = 1.20$). The original coding of the items indicated that a higher numerical value corresponded to a greater degree of cancer worry.

Smoking status (SS), also defined as smoking behavior (Anderson et al., 2022), was the product of two responses to “Have you smoked at least 100 cigarettes in your entire life?” and “How often do you now smoke cigarettes?”, creating a composite variable that reflects smoking behavior with three categories: 1 = current, 2 = former, and 3 = never. We recoded it as a binary variable (1 = current smoker; 0 = current nonsmoker, including never-smokers and former smokers; $M = 0.11$, $SD = 0.32$; Kiviniemi et al., 2011). Current smokers had smoked at least 100 cigarettes in their lifetime and reported smoking on some or every day. Former smokers had smoked at least 100 cigarettes but were not smoking at the time, while never smokers had not reached the 100-cigarette threshold.

Demographics included age, gender (1 = female, 0 = male), education (1 = *less than 8 years of schooling* to 7 = *postgraduate*), and political viewpoint (1 = *very liberal* to 7 = *very conservative*).

Analytical tools and approach

We conducted data analysis using SPSS (Version 29). First, descriptive analyses were performed to summarize the characteristics of each variable. Second, SPSS macro PROCESS (Hayes, 2017) Model 4 was applied to examine the mediation effect of CW on the association between SMUH and STCP, after controlling for demographics. Mediation effects were evaluated using bootstrapping with 5,000 samples and 95% confidence intervals (CI). Additionally, PROCESS Model 14 was used to explore the second stage moderated mediation model to investigate how SS moderates the relationship between CW and STCP (Hayes, 2015, 2018). We illustrate the mediation model (PROCESS Model 4) with Equation (1) and Equation (2),

$$CW = b_0 + b_1SMUH + b_2Gender + b_3Age + b_4Edu + b_5Pol + e \quad (1)$$

$$STCP = c_0 + c_1SMUH + c_2CW + c_3Gender + c_4Age + c_5Edu + c_6Pol + e \quad (2)$$

and demonstrate the second stage moderated mediation model (PROCESS Model 14) with Equation (3).

$$STCP = d_0 + d_1SMUH + d_2CW + d_3SS + d_4CW \times SS + d_5Gender + d_6Age + d_7Edu + d_8Pol + e \quad (3)$$

To reduce overreliance on significance tests and p values, two effect size measures were applied, percentage coefficient (b_p) and percent contribution (c_p). These supplement familiar indicators such as p and β , providing a more comprehensive estimate of effect size (Li et al., 2024). The b_p coefficient represents a b coefficient where both the dependent and

Table 1. Percentage transformation of study variables ($N = 2,671$).

		Non-01 natural scales (n_s)				Conceptual range		0–1% scales (p_s)			
		Min	Max	Mean	SD	Min	Max	Min	Max	Mean	SD
1.	Support for tobacco control policies	1	5	3.79	0.92	1	5	0	1	0.70	0.23
2.	Social media use for health	0	1	0.20	0.27	0	1	0	1	0.20	0.27
3.	Cancer worry	1	5	2.89	1.20	1	5	0	1	0.47	0.30
4.	Smoking status	1	3	2.51	0.69	0	1	0	1	0.11	0.32
5.	Gender	0	1	0.61	0.49	0	1	0	1	0.61	0.49
6.	Age	18	104	57.48	16.54	0	100	0.18	1.04	0.57	0.17
7.	Education	1	7	5.06	1.56	1	7	0	1	0.68	0.26
8.	Political viewpoint	1	7	4.14	1.63	1	7	0	1	0.52	0.27

independent variables are linearly transformed to a percentage scale (ranging from 0 to 1). The transformation follows Equation (4):

$$s_p = (s_{os} - s_{cn}) / (s_{cx} - s_{cn}) (0 \leq s_p \leq 1) \quad (4)$$

Where s_p is the percentage score after transformation, s_{os} is the original score, s_{cx} is the conceptual maximum on the original scale, and s_{cn} is the conceptual minimum on the original scale. Table 1 shows the transformation from the normal scale to the percentage scale. The percent contribution (c_p) measures the contribution of each pathway in the mediation model to the total effect of X on Y . Table 2 provides a detailed breakdown of the c_p calculations.

Results

Preliminary analyses

Table 3 presents the demographic characteristics of the respondents. On average, respondents were 57.48 years old, with a higher proportion of females (60.8%) compared to males (38.8%). Additionally, 67.7% of respondents reported having a college education or more. As indicated in Table 4, most of the key variables were significantly correlated. Better-educated individuals were more likely to use social media for health-related communication than those with less education ($r = .137$, $p < .001$). Liberals reported higher STCP ($r = -.058$, $p < .01$), which is consistent with prior findings (Shete et al., 2021). Females ($r = .112$, $p < .001$) and older people ($r = .096$, $p < .001$) showed a similar tendency toward higher STCP. Therefore, these variables were included as covariates to reduce potential confounding. Moreover, SMUH was positively associated with STCP ($r = .063$, $p < .01$), supporting the results of previous research (Hefler et al., 2013; Jackson et al., 2021).

Mediation testing

Table 5 and Figure 2 summarize the main findings, which are discussed below based on the hypotheses and research questions. Hypothesis 1 predicted that SMUH would be positively associated with STCP. As depicted in Table 5, the direct effect was positive and statistically significant ($b_p = 0.063$, $p < .001$). Hypothesis 2 posited that SMUH would have an indirect effect on STCP, mediated by CW. As shown in Table 5, SMUH was positively associated with CW ($b_p = 0.051$, $p = .034$) and CW was positively associated with STCP ($b_p = 0.060$, $p < .001$). Hypothesis 3 predicted a positive total effect of SMUH on STCP, which was supported by a significant positive total effect without controlling the mediator CW ($b_p = 0.066$, $p < .001$).

Key goodness-of-fit indicators were also examined. While the model explained a modest proportion of variance ($R^2 = .042$ for Equation II in Table 5), this was a notable improvement over Equation I ($R^2 = .008$). Additionally, the inclusion of demographic controls did not alter the significance or direction of the main effects, confirming the robustness of the relationships. The significant pathways support the theoretical role of CW as a mediator.

Research Question 1 addressed the type of mediation. The mediation model of this study, as proven above, is *complementary*. As illustrated in Tables 5 and 6, the direct effect ($b_p = 0.063$, $p < .001$) and the indirect effect ($b_p = 0.004$, 95% CI [0.000, 0.008]) exhibit the same positive direction and are statistically supported (X. Zhao et al., 2010).

Research Question 2 proposed a novel approach to interpreting effect sizes within a standardized 0–1% scale, where 0 and 1 represent the theoretically defined minimum

Table 2. Percent contribution to total X on Y effect (c_p).

Indicator	Equation	Range	Eq
Percent contribution of total effect to total effect (c)	$c_p(c) = b_p(c) / b_p(c) $	$c_p(c) = 1$ or $c_p(c) = -1$	1
Percent contribution of indirect effect (ab) to total effect (c)	$c_p(ab) = b_p(ab) / b_p(c) $	$-\infty < c_p(ab) < \infty$	2
Percent contribution of direct & remainder effect (d) to total effect (c)	$c_p(d) = b_p(d) / b_p(c) $	$-\infty < c_p(d) < \infty$	3
Percent contribution of 1 st -leg effect (a) to total effect (c)	$c_p(a) = (b_p(a)) / (b_p(a) + b_p(b)) \times c_p(ab)$	$-\infty < c_p(a) < \infty$	4
Percent contribution of 2 nd -leg effect (b) to total effect (c)	$c_p(b) = (b_p(b)) / (b_p(a) + b_p(b)) \times c_p(ab)$	$-\infty < c_p(b) < \infty$	5

Note: Before parenthesis: c_p , percent contribution to total effect; c , b_p , percentage coefficient. Within parenthesis: (a), first leg of the indirect path; (b), second leg of the indirect path; (ab), indirect path; (d), direct & remainder path; (c), total effect.

Table 3. Descriptive statistics of the dependent, independent, mediating, moderating, and controlling variables ($n = 2671$).

Dependent variable	
Support for Tobacco Control Policies (Five levels 1 ~ 5, Mean $\pm$ SD)	3.79 $\pm$ 0.92
Independent variable	
Social Media Use for Health (Two levels 0 ~ 1, Mean $\pm$ SD)	0.20 $\pm$ 0.27
Mediation variable	
Cancer Worry (Five levels 1 ~ 5, Mean $\pm$ SD)	2.89 $\pm$ 1.20
Moderating variable	
Smoking Status (Two levels 0 ~ 1, Mean $\pm$ SD)	0.11 $\pm$ 0.32
Sociodemographic controls	
Age (years, Mean $\pm$ SD)	57.48 $\pm$ 16.54
Gender (n. %)	
Male	1036 (38.8)
Female	1624 (60.8)
Education (n. %)	
Less than 8 years	32 (1.2)
8 through 11 years	111 (4.2)
12 years or completed high school	467 (17.5)
Post high school training other than college (vocational or technical)	190 (7.1)
Some college	580 (21.7)
College graduate	710 (26.6)
Post-graduate	518 (19.4)
Political Viewpoints (n. %)	
Very liberal	152 (5.7)
Liberal	323 (12.1)
Somewhat liberal	250 (9.4)
Moderate	831 (31.1)
Somewhat conservative	292 (10.9)
Conservative	410 (15.4)
Very conservative	195 (7.3)

Table 4. Zero-order Pearson correlations.

Variables	1	2	3	4	5	6	7	8
1. STCP	—							
2. SMUH	.063**	—						
3. CW	.77***	.066***	—					
4. SS	.138***	.067***	-.097***	—				
5. Gen	.112***	.120***	.013	.086***	—			
6. Age	.096***	-.286***	-.050*	-.098***	-.040*	—		
7. Edu	.020	.137***	-.026	.162***	-.028	-.198***	—	
(1) Pol	-.058**	-.081***	-.052*	.023	-.075***	.201***	-.135***	—

Note: STCP = support for tobacco control policies, SMUH = social media use for health, CW = cancer worry, SS = smoking status, Gen = gender (female), Age = age, Edu = education, Pol = political viewpoints. * $p < .05$; ** $p < .01$; *** $p < .001$.

Table 5. A product-and-process analysis of SMUH on STCP mediated by CW moderated by SS.

Right: Equation ID	Mediation Analysis		Moderation Analysis	Product Analysis
	Equation I	Equation II	Equation III	Equation IV
Right: Dependent Variable	CW	STCP	STCP	STCP
1. Intercept	0.568 ***	0.521 ***	0.562 ***	0.554 ***
2. Gender	0.002 (.013)	0.049 (.010) ***	0.046 (.010) ***	0.049 (.010) ***
3. Age	-0.079 (.040)*	0.211 (.030) ***	0.200 (.030) ***	0.206 (.030) ***
4. Edu	-0.056 (.025) *	0.025 (.019)	0.008 (.019)	0.022 (.019)
5. Pol	-0.043 (.023)	-0.059 (.018) ***	-0.066 (.018) ***	-0.061 (.018) ***
6. SMUH	0.051 (.024) *	0.063 (.018) ***	0.054 (.018) **	0.066 (.018) ***
7. CW		0.060 (.016) ***	0.050 (.017) **	
8. SS			-0.183 (.030) ***	
9. CW $\times$ SS			0.152 (.047) **	
Total r^2	0.008**	0.042***	0.064***	0.036***

Note. STCP = support for tobacco control policies, SMUH = social media use for health, CW = cancer worry, SS = smoking status, Gen = gender (female), Age = age, Edu = education, Pol = political viewpoints. All cells are b_p coefficients. * $p < .05$; ** $p < .01$; *** $p < .001$.

and maximum values of each variable. This standardization facilitates intuitive interpretation of associations between variables using effect sizes. For instance, as depicted in

Table 5, the effect size of SMUH on CW equals 0.051, indicating that a unit increase in SMUH is associated with a 5.1% increase in CW.

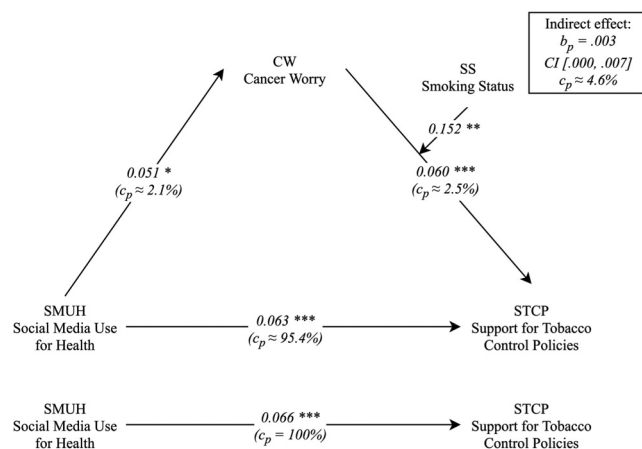


Figure 2. Effect of SMU on STCP mediated by CW and moderated by SS.

Moderation testing

Hypothesis 4 postulated the positive moderation effect of SS in the association between CW and STCP. As shown in Table 5, the interaction term (CW $\times$ SS) was statistically significant ($b_p = 0.152$, $p < .01$). Adding the interaction term improved the model fit, explaining 6.4% of the variance in STCP (Equation III, $R^2 = .064$). The negative coefficient for SS ($b_p = -0.183$,

$p < .001$) further indicated that the relationship between CW and STCP differs between smokers and nonsmokers. Hypothesis 5 predicted that SS positively moderates the indirect path between SMUH and STCP through CW. The moderated mediation index ($b_p = 0.008$, 95% CI [0.000, 0.020]) in Table 6 was significant, confirming a second-stage moderated mediation.

Figure 3 depicts the moderation effects on the second stage of mediation (the b path). SS positively moderates the relationship between CW and STCP. Notably, in the subgroup with a family cancer history, smokers exhibited a steeper increase in STCP as CW intensified compared to nonsmokers, even though nonsmokers had higher baseline support.

Summary and discussion

This study aimed to examine a complementary mediation mechanism through CW in the relationship between SMUH and STCP among U.S. adults with a family history of cancer. The findings underscore the direct and indirect effects of SMUH on STCP, positively mediated by CW and positively moderated by SS.

Table 6. Regressions testing the moderated mediation model.

	b	BootSE	95% CI
Mediation model (indirect effect)			
SMUH $\rightarrow$ CW $\rightarrow$ STCP	0.004	0.002	[0.000, 0.008]
Moderated mediation model			
Index of moderated mediation (SS $\rightarrow$ indirect path)	0.008	0.005	[0.000, 0.020]

Note: STCP = support for tobacco control policies, SMUH = social media use for health, CW = cancer worry, SS = smoking status. Due to rounding, the reported CI [0.000, 0.008] corresponds to the exact interval [0.0004, 0.0078], which excludes zero. Similarly, the reported CI [0.000, 0.020] corresponds to [0.0002, 0.0195], which also excludes zero. Thus, both effects are significant at $p < .05$.

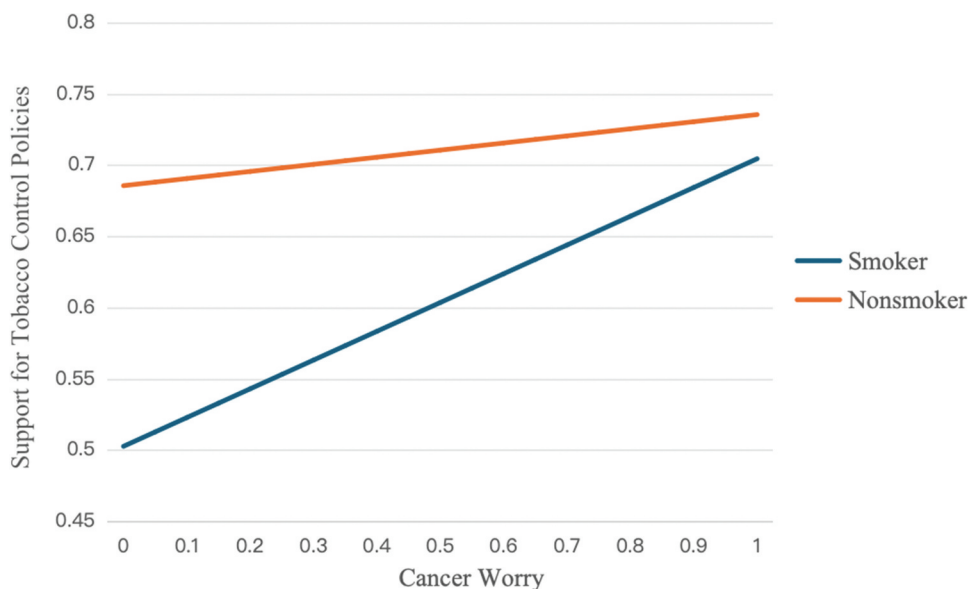


Figure 3. Moderation effect of smoking status. Note: Cancer worry and support for tobacco control policies are displayed on 0–1% scales; 0 denotes the conceptual minimum and 1 denotes the conceptual maximum.

Theoretical implications

This research offers several theoretical contributions. Using the S-O-R model, this study demonstrates how SMUH is positively associated with STCP through complementary mediation. SMUH serves as an external stimulus. This stimulus is positively associated with an organismic state, namely CW, which in turn is positively associated with a behavioral response: STCP. This framework provides a theoretically grounded account of health communication interventions delivered via social media and can further predict engagement in health-promoting behaviors.

Unique to this investigation is the exploration of how SS moderates the relationship between CW and STCP. The results reveal a nuanced insight: among individuals with a family history of cancer, smokers show a steeper increase in STCP as CW intensifies, compared to nonsmokers.

Practical implications

One of the practical implications of this research is the positive association between SMUH and STCP. This suggests that health information obtained from social media is linked to more favorable attitudes toward tobacco restrictions. Despite the prevalence of misinformation, social media remains an effective tool for helping individuals acquire more accurate knowledge about smoking. The widespread use of social media may therefore contribute to the formation of public health knowledge.

Enlightened by the S-O-R model, policymakers may consider reinforcing tobacco control through both social media approaches and the arousal of awareness and emotional responses regarding the carcinogenic nature of tobacco. This dual approach is especially crucial for the subgroup with a family cancer history, who may exhibit heightened sensitivity to cancer-related stimuli. Specifically, beyond merely disseminating information, social media platforms offer powerful tools to (a) provide objective and precise information about the health risks due to smoking, (b) implement real-time surveillance to identify and counter tobacco promotion tactics, and (c) foster community engagement to amplify the public perception of cancer risk from tobacco. By effectively utilizing these capabilities, policymakers could not only enhance public awareness of cancer risk but also directly cultivate stronger public support for evidence-based tobacco control policies.

Importantly, this study measured behaviors such as sharing health information and participating in online forums, representing active user engagement rather than passive exposure to stimuli. Future interventions should therefore leverage these engagement behaviors to move beyond the traditional S-O-R framework. For instance, policymakers could design interactive campaigns that encourage users to share personal stories about the impact of smoking on individuals with a family cancer history, thereby transforming passive viewers into active advocates. Additionally, facilitating moderated online forums where individuals with a family cancer history can discuss tobacco control policies may amplify the cancer worry mechanism identified in this study, turning it into a collective call for action. By harnessing these forms of

participation, social media interventions can shift from merely providing information (stimulus) to cultivating a supportive community that actively promotes policy change (response).

Limitations

This study has several limitations. First, the application of the S-O-R framework within a cross-sectional survey design inherently constrains causal inference. While the model posits a sequence, the survey data capture associations at a single point in time. This makes it difficult to definitively establish the temporal ordering or directionality of effects from social media use for health (stimulus) to cancer worry (organism), and to support for tobacco control policies (response). Accordingly, these findings should be interpreted as revealing statistical associations that are consistent with the S-O-R theoretical mechanism. Future studies should employ longitudinal designs or experimental manipulations to rigorously test the causal arrows proposed by the S-O-R model. Such designs would permit stronger inferences regarding whether exposure to health information on social media precedes changes in cancer worry, and whether these changes in turn drive shifts in tobacco control policy support.

Second, the measurement of exposure to general health information rather than cancer-specific content may yield different affective outcomes than those elicited by cancer-related material, which often elicits distinct emotions. Therefore, the findings may not fully capture the specific impact of cancer-focused social media exposure. Future research should replicate these findings using measures tailored to cancer-specific content.

Third, the low response rate was a common drawback of mailed surveys and may result in sample bias. However, this is less of a concern in the present design, as individual characteristics typically associated with nonresponse do not lead to systematic missing data or at least do not directly bias the main variables of interest. Nevertheless, future studies should collect data from more comprehensive sources to overcome this flaw.

Conclusion

In summary, this study advances understanding of how health-related social media use shapes public support for tobacco control. It also introduces a theoretical model of cancer worry mediation moderated by smoking status within the S-O-R framework. Furthermore, the findings provide practical guidance for policymakers designing targeted tobacco control interventions.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

- Amrhein, V., Greenland, S., & McShane, B. (2019). Scientists rise up against statistical significance. *Nature*, 567(7748), 305–307. <https://doi.org/10.1038/d41586-019-00857-9>
- Anderson, S. L., Livingston, M. D., Higgins, K. A., & McBride, C. M. (2022). Predictors of discussing lung cancer screening with a health care provider among current and former smokers in HINTS: A secondary data analysis. *Cancer Control*, 29. <https://doi.org/10.1177/10732748221130567>
- Bakhturidze, G. D., Mittelmarm, M. B., Aarø, L. E., & Peikrishvili, N. T. (2013). Attitudes towards smoking restrictions and tobacco advertisement bans in Georgia. *BMJ Open*, 3(11), e003461. <https://doi.org/10.1136/bmjopen-2013-003461>
- Bialous, S. A., Kaufman, N., & Sarna, L. (2003). Tobacco control policies. *Seminars in Oncology Nursing*, 19(4), 291–300. <https://doi.org/10.1053/j.soncn.2003.08.007>
- Boderie, N. W., Sheikh, A., Lo, E., Sheikh, A., Burdorf, A., van Lenthe, F. J., Mölénberg, F. J., & Been, J. V. (2023). Public support for smoke-free policies in outdoor areas and (semi-) private places: A systematic review and meta-analysis. *EClinicalMedicine*, 59, 101982. <https://doi.org/10.1016/j.eclinm.2023.101982>
- Braun, L. A., Zomorodbakhsch, B., Keinki, C., & Huebner, J. (2019). Information needs, communication and usage of social media by cancer patients and their relatives. *Journal of Cancer Research and Clinical Oncology*, 145(7), 1865–1875. <https://doi.org/10.1007/s00432-019-02929-9>
- Brown, E. R., McCarthy, W. J., Marcus, A., Baker, D., Froines, J. R., Dellenbaugh, C., & McQuiston, T. (1988). Workplace smoking policies: Attitudes of union members in a high-risk industry. *Journal of Occupational Medicine*, 30(4), 312–320.
- Butterworth, A. S., Higgins, J. P., & Pharoah, P. (2006). Relative and absolute risk of colorectal cancer for individuals with a family history: A meta-analysis. *European Journal of Cancer*, 42(2), 216–227. <https://doi.org/10.1016/j.ejca.2005.09.023>
- Cavazos-Rehg, P., Li, X., Kasson, E., Kaiser, N., Borodovsky, J. T., Grucza, R., Chen, L.-S., & Bierut, L. J. (2021). Exploring how social media exposure and interactions are associated with ENDS and tobacco use in adolescents from the PATH study. *Nicotine & Tobacco Research*, 23(3), 487–494. <https://doi.org/10.1093/ntr/ntaa113>
- Chaloupka, F. J., Yurekli, A., & Fong, G. T. (2012). Tobacco taxes as a tobacco control strategy. *Tobacco Control*, 21(2), 172–180. <https://doi.org/10.1136/tobaccocontrol-2011-050417>
- Chen, L., & Kaphingst, K. (2010). Risk perceptions and family history of lung cancer: Differences by smoking status. *Public Health Genomics*, 14(1), 26–34. <https://doi.org/10.1159/000294151>
- Chen, Y., & Yang, Q. (2017). How do cancer risk perception, benefit perception of quitting, and cancer worry influence quitting intention among current smokers: A study using the 2013 HINTS. *Journal of Substance Use*, 22(5), 555–560. <https://doi.org/10.1080/14659891.2016.1271033>
- Chou, W.-Y. S., Gaysynsky, A., Trivedi, N., & Vanderpool, R. C. (2021). Using social media for health: National data from HINTS 2019. *Journal of Health Communication*, 26(3), 184–193. <https://doi.org/10.1080/10810730.2021.1903627>
- Chu, K. H., Allem, J. P., Cruz, T. B., & Unger, J. B. (2017). Vaping on Instagram: Cloud chasing, hand checks and product placement. *Tobacco Control*, 26(5), 575–578. <https://doi.org/10.1136/tobaccocontrol-2016-053052>
- Dillard, A. J., McCaul, K. D., & Klein, W. M. (2006). Unrealistic optimism in smokers: Implications for smoking myth endorsement and self-protective motivation. *Journal of Health Communication*, 11(S1), 93–102. <https://doi.org/10.1080/10810730600637343>
- Dobbs, P. D., Boykin, A. A., Ezike, N., Myers, A. J., Colditz, J. B., & Primack, B. A. (2023). Twitter sentiment about the US federal tobacco 21 law: Mixed methods analysis. *JMIR Formative Research*, 7, e50346. <https://doi.org/10.2196/50346>
- Donaldson, S. I., Dormanesh, A., Majmundar, A., Pérez, C., Lopez, H., Saghian, M., Beard, T. A., Unger, J. B., & Allem, J. P. (2024). Examining the peer-reviewed literature on tobacco-related social media data: Scoping review. *Nicotine & Tobacco Research*, 26(4), 413–420. <https://doi.org/10.1093/ntr/ntad186>
- Donaldson, S. I., Dormanesh, A., Perez, C., Majmundar, A., & Allem, J. P. (2022). Association between exposure to tobacco content on social media and tobacco use: A systematic review and meta-analysis. *JAMA Pediatrics*, 176(9), 878–885. <https://doi.org/10.1001/jamapediatrics.2022.2223>
- Duke, J. C., Alexander, T. N., Zhao, X., Delahanty, J. C., Allen, J. A., MacMonegle, A. J., & Farrelly, M. C. (2015). Youth's awareness of and reactions to the real cost national tobacco public education campaign. *PLOS ONE*, 10(12), e0144827. <https://doi.org/10.1371/journal.pone.0144827>
- Duke, J. C., Davis, K. C., Alexander, R. L., MacMonegle, A. J., Frazee, J. L., Rodes, R. M., & Beistle, D. M. (2015). Impact of a US antismoking national media campaign on beliefs, cognitions and quit intentions. *Health Education Research*, 30(3), 466–483. <https://doi.org/10.1093/her/cyv017>
- Emery, S. L., Vera, L., Huang, J., & Szczypka, G. (2014). Wanna know about vaping? Patterns of message exposure, seeking and sharing information about e-cigarettes across media platforms. *Tobacco Control*, 23(3), iii17–iii25. <https://doi.org/10.1136/tobaccocontrol-2014-051648>
- Fakfare, P., & Sangpikul, A. (2024). Resident perceptions towards COVID-19 public policies for tourism reactivation: The case of Thailand. *Journal of Policy Research in Tourism, Leisure and Events*, 16(4), 785–805. <https://doi.org/10.1080/19407963.2022.2076689>
- Fan, M., Ezeudoka, B. C., & Qalati, S. A. (2024). Exploring the resistance to e-health services in Nigeria: An integrative model based upon the theory of planned behavior and stimulus-organism-response. *Humanities and Social Sciences Communications*, 11(1), Article 571. <https://doi.org/10.1057/s41599-024-03090-6>
- Farrelly, M. C., Nonnemaker, J., Davis, K. C., & Hussin, A. (2009). The influence of the national truth® campaign on smoking initiation. *The American Journal of Preventive Medicine*, 36(5), 379–384. <https://doi.org/10.1016/j.amepre.2009.01.019>
- Feng, R. M., Zong, Y. N., Cao, S. M., & Xu, R. H. (2019). Current cancer situation in China: Good or bad news from the 2018 global cancer statistics? *Cancer Communications*, 39(1), Article 22. <https://doi.org/10.1186/s40880-019-0368-6>
- Ferrer, R. A., Portnoy, D. B., & Klein, W. M. (2013). Worry and risk perceptions as independent and interacting predictors of health protective behaviors. *Journal of Health Communication*, 18(4), 397–409. <https://doi.org/10.1080/10810730.2012.727954>
- Finney Rutten, L. J., Blake, K. D., Hesse, B. W., Augustson, E. M., & Evans, S. (2011). Illness representations of lung cancer, lung cancer worry, and perceptions of risk by smoking status. *Journal of Cancer Education*, 26(4), 747–753. <https://doi.org/10.1007/s13187-011-0247-6>
- Fotuhi, O., Fong, G. T., Zanna, M. P., Borland, R., Yong, H.-H., & Cummings, K. M. (2013). Patterns of cognitive dissonance-reducing beliefs among smokers: A longitudinal analysis from the International tobacco control (ITC) four country survey. *Tobacco Control*, 22(1), 52–58. <https://doi.org/10.1136/tobaccocontrol-2011-050139>
- Fu, S., Chen, X., & Zheng, H. (2021). Exploring an adverse impact of smartphone overuse on academic performance via health issues: A stimulus-organism-response perspective. *Behaviour & Information Technology*, 40(7), 663–675. <https://doi.org/10.1080/0144929X.2020.1716848>
- Gredner, T., Mons, U., Niedermaier, T., Brenner, H., & Soerjomataram, I. (2021). Impact of tobacco control policies implementation on future lung cancer incidence in Europe: An international, population-based modeling study. *The Lancet Regional Health-Europe*, 4, 100074. <https://doi.org/10.1016/j.lanepe.2021.100074>
- Hamill, S., Turk, T., Murukutla, N., Ghamrawy, M., & Mullin, S. (2015). I 'like' MPOWER: Using Facebook, online ads and new media to mobilise tobacco control communities in low-income and middle-income countries. *Tobacco Control*, 24(3), 306–312. <https://doi.org/10.1136/tobaccocontrol-2012-050946>
- Hawkins, S. S., Chung-Hall, J., Craig, L., Fong, G. T., Borland, R., Cummings, K. M., Levy, D., & Hitchman, S. C. (2020). Support for

- minimum legal sales age laws set to age 21 across Australia, Canada, England, and United States: Findings from the 2018 ITC four country smoking and vaping survey. *Nicotine & Tobacco Research*, 22(12), 2266–2270. <https://doi.org/10.1093/ntr/ntaa119>
- Hayes, A. F. (2015). An index and test of linear moderated mediation. *Multivariate Behavioral Research*, 50(1), 1–22. <https://doi.org/10.1080/00273171.2014.962683>
- Hayes, A. F. (2017). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. The Guilford Press.
- Hayes, A. F. (2018). Partial, conditional, and moderated moderated mediation: Quantification, inference, and interpretation. *Communication Monographs*, 85(1), 4–40. <https://doi.org/10.1080/03637751.2017.1352100>
- Hefler, M., Freeman, B., & Chapman, S. (2013). Tobacco control advocacy in the age of social media: Using Facebook, Twitter and change. *Tobacco Control*, 22(3), 210–214. <https://doi.org/10.1136/tobaccocontrol-2012-050721>
- Henderson, K. C., Loud, E. E., Duong, H. T., Reynolds, R. M., Yang, B., Ntansah, C. A., Ashley, D. L., Thrasher, J. F., & Popova, L. (2022). Perceptions of nicotine reduction policy in the United States: A qualitative study. *Nicotine & Tobacco Research*, 24(9), 1422–1429. <https://doi.org/10.1093/ntr/ntac071>
- Hoffman-Goetz, L., Meissner, H., & Thomson, M. (2009). Literacy and cancer anxiety as predictors of health status: An exploratory study. *Journal of Cancer Education*, 24(3), 218–224. <https://doi.org/10.1080/08858190902910871>
- Huang, J., Kornfield, R., & Emery, S. L. (2016). 100 million views of electronic cigarette YouTube videos and counting: Quantification, content evaluation, and engagement levels of videos. *Journal of Medical Internet Research*, 18(3), e67. <https://doi.org/10.2196/jmir.4265>
- Huo, J., Desai, R., Hong, Y.-R., Turner, K., Mainous, A. G., III, & Bian, J. (2019). Use of social media in health communication: Findings from the health information National Trends survey 2013, 2014, and 2017. *Cancer Control*, 26(1). <https://doi.org/10.1177/1073274819841442>
- Jackson, M., Brennan, L., & Parker, L. (2021). The public health community's use of social media for policy advocacy: A scoping review and suggestions to advance the field. *Public Health*, 198, 146–155. <https://doi.org/10.1016/j.puhe.2021.07.015>
- Jung, M. (2014). Moderating effects of media exposure between socioeconomic position and cancer worry. *Asian Pacific Journal of Cancer Prevention*, 15(14), 5845–5851. <https://doi.org/10.7314/apjcp.2014.15.14.5845>
- Kaufman, A. R., D'Angelo, H., Gaysynsky, A., Seidenberg, A. B., Vollinger, R. E., Jr., & Blake, K. D. (2022). Public support for cigarette pack pictorial health warnings among US adults: A cross-sectional analysis of the 2020 health information national Trends survey. *Nicotine & Tobacco Research*, 24(6), 924–928. <https://doi.org/10.1093/ntr/ntab263>
- Kiviniemi, M. T., Orom, H., & Giovino, G. A. (2011). Psychological distress and smoking behavior: The nature of the relation differs by race/ethnicity. *Nicotine & Tobacco Research*, 13(2), 113–119. <https://doi.org/10.1093/ntr/ntq218>
- Kong, G., Blake, K. D., & Romer, D. (2024). Advances in social media research to reduce tobacco use. *Nicotine & Tobacco Research*, 26(1), S1–S2. <https://doi.org/10.1093/ntr/ntad250>
- Kong, G., Laestadius, L., Vasey, J., Majmundar, A., Stroup, A. M., Meissner, H. I., Taleb, Z. B., Cruz, T. B., Emery, S. L., & Romer, D. (2024). Tobacco promotion restriction policies on social media. *Tobacco Control*, 33(3), 398–403. <https://doi.org/10.1136/tc-2022-057348>
- Lama, Y., Nan, X., & Quinn, S. C. (2022). General and health-related social media use among adults with children in the household: Findings from a national survey in the United States. *Patient Education and Counseling*, 105(3), 647–653. <https://doi.org/10.1016/j.pec.2021.07.003>
- Land, S. R., Baker, L., Twesten, J., Reyes-Guzman, C. M., & Kaufman, A. R. (2023). Smoking cessation and tobacco-related risk perceptions among people with and without a diagnosis of cancer. *Cancer Epidemiology, Biomarkers & Prevention*, 32(2), 266–273. <https://doi.org/10.1158/1055-9965.EPI-22-0651>
- Lee, P. N. (2018). Tar level of cigarettes smoked and risk of smoking-related diseases. *Inhalation Toxicology*, 30(1), 5–18. <https://doi.org/10.1080/08958378.2018.1443174>
- Li, M. M., Zhao, X., & Tuo, Z. T. (2024). Influence of social media on public support for cancer prevention policies. *Interdisciplinary Social Research Reports*, 1, 1–18. <https://www.acspublisher.org/index.php/isrr/article/view/28>
- Liang, Y., Zheng, X., Zeng, D. D., Zhou, X., Leischow, S. J., & Chung, W. (2015). Exploring how the tobacco industry presents and promotes itself in social media. *Journal of Medical Internet Research*, 17(1), e3665. <https://doi.org/10.2196/jmir.3665>
- Liu, P. L. (2020). COVID-19 information seeking on digital media and preventive behaviors: The mediation role of worry. *Cyberpsychology, Behavior and Social Networking*, 23(10), 677–682. <https://doi.org/10.1089/cyber.2020.0250>
- Liu, P. L., Ye, J. F., Ao, H. S., Sun, S., Zheng, Y., Li, Q., Feng, G. C., Wang, H., & Zhao, X. (2023). Effects of electronic personal health information technology on American women's cancer screening behaviors mediated through cancer worry: Differences and similarities between 2017 and 2020. *Digital Health*, 9. <https://doi.org/10.1177/20552076231185271>
- Mehrabian, A., & Russell, J. A. (1974). *An approach to environmental psychology*. The MIT Press.
- Molina, Y., Ceballos, R. M., Dolan, E. D., Albano, D., & McGregor, B. A. (2015). Perceived breast cancer risk and breast cancer worry among women with a family history of breast cancer: A new perspective on coping as a mediator. *Psycho-Oncology*, 24(1), 113–116. <https://doi.org/10.1002/pon.3587>
- Mumford, E. A., Pearson, J. L., Villanti, A. C., & Evans, W. D. (2017). Nicotine and e-cigarette beliefs and policy support among US smokers and nonsmokers. *Tobacco Regulatory Science*, 3(3), 293–305. <https://doi.org/10.18001/TRS.3.3.5>
- Munthali, G. N. C., Wu, X. L., Rizwan, M., Daru, G. R., & Shi, Y. (2021). Assessment of tobacco control policy instruments, status and effectiveness in Africa: A systematic literature review. *Risk Management and Healthcare Policy*, 14, 2913–2927. <https://doi.org/10.2147/RMHP.S311551>
- Onyeaka, H., Firth, J., Ajayi, K. V., Muoghalu, C., Holmes, K., Nkemjika, S., Adeolu, F., Anugwom, G., Eseaton, P. O., Onyeaka, N. C., Huffman, J., & Torous, J. (2023). Association between social media use and health promotion among individuals with depression and anxiety: Insights from the 2017–2020 Health Information National Trends survey. *Journal of Mood and Anxiety Disorders*, 1, 100006. <https://doi.org/10.1016/j.xjmad.2023.100006>
- Paoletti, L., Jardin, B., Carpenter, M. J., Cummings, K. M., & Silvestri, G. A. (2012). Current status of tobacco policy and control. *Journal of Thoracic Imaging*, 27(4), 213–219. <https://doi.org/10.1097/RTI.0b013e3182518673>
- Reis, E. S., & Postolache, O. A. (2019). Fear or humour in anti-smoking campaigns? Impact on perceived effectiveness and support for tobacco control policies. *Ciencia & Saude Coletiva*, 24(12), 4727–4738. <https://doi.org/10.1590/1413-812320182412.09322018>
- Rosenberg, M., Pettigrew, S., Wood, L., Ferguson, R., & Houghton, S. (2012). Public support for tobacco control policy extensions in western Australia: A cross-sectional study. *BMJ Open*, 2(2), e000784. <https://doi.org/10.1136/bmjopen-2011-000784>
- Semple, S., Dobson, R., O'Donnell, R., Abidin, E. Z., Tigova, O., Okello, G., & Fernández, E. (2022). Smoke-free spaces: A decade of progress, a need for more? *Tobacco Control*, 31(2), 250–256. <https://doi.org/10.1136/tobaccocontrol-2021-056556>
- Seven, M., Bağcıvan, G., Akyuz, A., & Bölükbaş, F. (2018). Women with family history of breast cancer: How much are they aware of their risk? *Journal of Cancer Education*, 33(4), 915–921. <https://doi.org/10.1007/s13187-017-1226-3>
- Shete, S. S., Yu, R., & Shete, S. (2021). Political ideology and the support or opposition to United States tobacco control policies. *JAMA Network Open*, 4(9), e2125385. <https://doi.org/10.1001/jamanetworkopen.2021.25385>

- Strong, D. R., Pierce, J. P., Pulvers, K., Stone, M. D., Villaseñor, A., Pu, M., Dimofte, C. V., Leas, E. C., Oratowski, J., Brighton, E., Hurst, S., Kealey, S., Chen, R., & Messer, K. (2021). Effect of graphic warning labels on cigarette packs on US smokers' cognitions and smoking behavior after 3 months: A randomized clinical trial. *JAMA Network Open*, 4(8), e2121387. <https://doi.org/10.1001/jamanetworkopen.2021.21387>
- Sultan, P., Wong, H. Y., & Azam, M. S. (2021). How perceived communication source and food value stimulate purchase intention of organic food: An examination of the stimulus-organism-response (SOR) model. *Journal of Cleaner Production*, 312, 127807. <https://doi.org/10.1016/j.jclepro.2021.127807>
- Tsay, S. L., Wang, J. Y., Lee, Y. H., & Chen, Y. J. (2020). Fear of recurrence: A mediator of the relationship between physical symptoms and quality of life in head and neck cancer patients. *European Journal of Cancer Care*, 29(4), e13243. <https://doi.org/10.1111/ecc.13243>
- Tyc, V. L., Puleo, E., Emmons, K., De Moor, J. S., & Ford, J. S. (2013). Smoking restrictions among households of childhood and young adult cancer survivors: Implications for tobacco control efforts. *Journal of Adolescent and Young Adult Oncology*, 2(1), 17–24. <https://doi.org/10.1089/jayao.2012.0019>
- Vallone, D., Cantrell, J., Bennett, M., Smith, A., Rath, J. M., Xiao, H., Greenberg, M., & Hair, E. C. (2018). Evidence of the impact of the truth FinishIt campaign. *Nicotine & Tobacco Research*, 20(5), 543–551. <https://doi.org/10.1093/ntr/ntx119>
- Vance, K., Howe, W., & Dellavalle, R. P. (2009). Social internet sites as a source of public health information. *Dermatologic Clinics*, 27(2), 133–136. <https://doi.org/10.1016/j.det.2008.11.010>
- Winston, S. (2021). Health information national trends survey (Hints. Gov). *Medical Reference Services Quarterly*, 40(2), 215–223. <https://doi.org/10.1080/02763869.2021.1912575>
- World Health Organization. (2026a, July 3). *Cancer*. <https://www.who.int/news-room/fact-sheets/detail/cancer>
- World Health Organization. (2026b, June 26). *Tobacco and nicotine*. <https://www.who.int/news-room/fact-sheets/detail/tobacco>
- Yoon, P. W., Scheuner, M. T., Peterson-Oehlke, K. L., Gwinn, M., Faucett, A., & Khoury, M. J. (2002). Can family history be used as a tool for public health and preventive medicine? *Genetics in Medicine*, 4(4), 304–310. <https://doi.org/10.1097/00125817-200207000-00009>
- Zhang, L., Ao, S. H., Ye, J. F., & Zhao, X. (2024). How does health communication on social media influence e-cigarette perception and use? A trend analysis from 2017 to 2020. *Addictive Behaviors*, 149, 107875. <https://doi.org/10.1016/j.addbeh.2023.107875>
- Zhang, X., Liu, Y., Qin, Z., Ye, Z., & Meng, F. (2023). Understanding the role of social media usage and health self-efficacy in the processing of COVID-19 rumors: A SOR perspective. *Data and Information Management*, 7(2), 100043. <https://doi.org/10.1016/j.dim.2023.100043>
- Zhao, X., Li, D. M., Lai, Z. Z., Liu, P. L., Ao, S. H., & You, F. (2024). Percentage coefficient (bp)-effect size analysis (theory paper 1). *arXiv Preprint arXiv: 2404.19495*. <https://doi.org/10.48550/arXiv.2404.19495>
- Zhao, X., Lynch, J. G., Jr., & Chen, Q. (2010). Reconsidering Baron and Kenny: Myths and truths about mediation analysis. *Journal of Consumer Research*, 37(2), 197–206. <https://doi.org/10.1086/651257>
- Zhao, Y., & Zhang, J. (2017). Consumer health information seeking in social media: A literature review. *Health Information and Libraries Journal*, 34(4), 268–283. <https://doi.org/10.1111/hir.12192>